Project Proposal: Extractive QA with Confidence Calibration

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1. Approach. We fine-tune BERT on SQuAD v1.1 to predict answer start and end positions. Our central question is whether a *learned* confidence estimate is better calibrated than the standard *softmax* confidence ($P_{start} \times P_{end}$ of the predicted span), which may be miscalibrated.

Main method: a [CLS]-based head, $p = \sigma(w \cdot h_{CLS} + b)$, trained jointly with $L = L_{QA} + \lambda \cdot L_{conf}$. L_{conf} is binary cross-entropy against y, where y = 1 if the predicted span exactly matches a gold answer, else 0 (computed without gradients). We ablate detaching h_{CLS} from BERT.

Hypothesis: h_{CLS} does not see the predicted span, so it likely captures question-level difficulty rather than span-level correctness. We test a span-conditioned head that concatenates h_{CLS} with mean-pooled hidden states over the predicted span.

Calibration: post-hoc temperature scaling is applied to both softmax and learned confidence, with a separate T fit for each signal.

2. Evaluation. We train on train-v1.1.json and evaluate on dev-v1.1.json with the official script (EM, F1), and measure calibration with Expected Calibration Error (ECE). T is fit on a held-out 10% of the train. Comparison: (1) BERT QA baseline; (2) raw softmax confidence; (3) softmax + temperature scaling; (4) [CLS] head; (5) [CLS] head + temperature scaling; (6) span-conditioned head + temperature scaling. Key comparisons are rows 3, 5 and 6. *Stretch goals (only if the core is complete):* ECE and mean confidence by question type, passage length, answer length and baseline F1 (difficulty analysis); confidence-aware top-k span selection.

3. Data and External Code. No additional data or annotation; all experiments use SQuAD v1.1 train and dev. We use Hugging Face Transformers, PyTorch and the official SQuAD evaluation script.

4. Related Work and Uniqueness. Temperature scaling and ECE follow Guo et al. [1]. Calibration of pre-trained BERT/RoBERTa follows Desai & Durrett [2], and training a calibrator to predict QA errors follows Kamath et al. [3]. Our contribution is a systematic comparison of learned versus softmax confidence on SQuAD, a test of whether conditioning on the predicted span helps, and an analysis of where calibration fails.

5. Work Breakdown. M1: BERT baseline and training. M2: preprocessing and data pipeline. M3: [CLS] and span-conditioned heads, loss. M4: temperature scaling and ECE. M5: evaluation, ablations, difficulty analysis, visualisation. All: experiment design, report, presentation.

6. Timeline. Wk 8: preprocessing, BERT baseline, softmax confidence and ECE. Wk 9: baseline EM/F1; [CLS] head and loss; fix held-out split. Wk 10: temperature scaling; span-conditioned head. Wk 11: ablations (detach, EM vs. F1 target), reliability diagrams, error analysis; stretch goals if on track. Wk 12: report and presentation. Wk 13: buffer and submission.

References

[1] Guo, Pleiss, Sun, Weinberger. On Calibration of Modern Neural Networks. ICML 2017. arxiv.org/abs/1706.04599

[2] Desai, Durrett. Calibration of Pre-trained Transformers. EMNLP 2020. arxiv.org/abs/2003.07892

[3] Kamath, Jia, Liang. Selective Question Answering under Domain Shift. ACL 2020. arxiv.org/abs/2006.09462